

CST4812 - Natural Language Processing

Final Project

Project Description:

The final project is open ended. You may complete this either as an individual or as a group with up to two total members.

The purpose of the project is to use PyTorch to implement any project involving an NLP task and a neural network.

For most tasks, I would prefer that you use some variation of a recurrent neural network or transformers, but I am open to other sorts of architectures if you think something else makes sense for your project.

The task can be something that we have covered, or will cover, in class, but it does not have to be. You should test your implementation on a dataset and report the results. You are allowed to create a dataset, but it will be much easier if you find one. Kaggle, for example, has plenty of open source datasets for NLP problems for you to leverage.

Some possible project ideas are:

- Neural Machine Translation
- Sentiment Analysis
- POS tagging
- Text Generation
- Text Summarization
- Question-Answering Systems

Feel free to choose a task from the above list, or explore something not mentioned. You may notice that some of these tasks resemble other assignments from this semester. You are free to use your past assignments as a resource, but I expect you to submit something new with this project. I do not just want to see a past assignment resubmitted as your submission for this project.

When you have completed your project, submit a SHORT writeup (one to two pages) explaining:

- What task did you work on?
- What dataset did you use?
- How was the dataset split into training, tuning, and testing portions?
- What architectures and other variables did you experiment with?
- Summarize the results of the things that were varied.
- What were the settings used for the final network?
- What were the final results for the test set?

Due Date: Tuesday May 26th, 11:59pm EST